

Problem 1: Build three piles. Take turns removing one or more counters from exactly one pile. Whoever takes the last counter wins. Play (1,2,3), then (1,2,4). Change who starts and play each again. Use the spaces for your piles.

Start	Who won?	One first move and reply
(1,2,3)		
(1,2,3)		
(1,2,4)		
(1,2,4)		

Problem 2: Replay one game. Choose a different first move and record the reply. Circle a start you would prefer to give your opponent.

Problem 3: Test these starts with counters. For each, try to leave two equal piles and one empty pile in a single move. Write a move if you find one, or record what you tried.

Start	Total	Move or attempts
(1,1,1)		
(1,1,2)		
(1,2,3)		
(2,4,6)		

Problem 4: Rebuild (1,2,3) after each trial. Your partner chooses each first move in this table. Find a reply leaving two equal piles and one empty pile; write the result.

Partner leaves	My reply leaves
(0,2,3)	
(1,0,3)	
(1,1,3)	
(1,2,0)	
(1,2,1)	
(1,2,2)	

Problem 5: Group each pile into bundles of 4, 2, and 1, using each size at most once. Record 1 if you use that bundle, 0 if you do not. The first row shows 5 counters grouped as $4+1$. Make and record the other piles with counters.

Pile	4	2	1	Bundle sum
5	1	0	1	$4+1$
1				
2				
3				
4				
6				
7				

Problem 6: Represent these two games using the same bundle records. Count the 1s in each column. Circle every odd count. Call a game balanced if all three counts are even; zero is even.

Pile	4	2	1	Pile	4	2	1
1				1			
2				2			
3				4			
Count				Count			

Problem 7: For (1,2,4), reduce just one pile so all column counts become even. Make that move with ordinary counters. Record the new piles.

Problem 8: For each start, write each pile’s 4/2/1 record. If all column counts are even, write “balanced.” Otherwise shrink one pile to make them even and record the new piles. Make each move with counters.

Start / pile	4	2	1	Move or balanced
2				
4				
6				
3				
4				
5				
1				
5				
7				
3				
5				
6				

Problem 9: Choose one balanced start above. Let your partner make a move. Record the new piles, then find and play a move that restores balance. Repeat with a different partner move.

Partner leaves	My reply leaves

Problem 10: Try this way to repair odd columns. Find the largest bundle column with an odd count. Choose a pile using that bundle. Keep its larger bundles, remove this bundle, and choose its smaller bundles to make every column even. Try (4,6,7), then (7,10,12). Record the starting bundles and the replacement pile. Check that it is smaller, then make the move with counters.

Pile	8	4	2	1	Replacement pile
4					
6					
7					
7					
10					
12					

Problem 11: Choose a third pile that balances (2,5,____), then (3,6,____). Build both games and check their columns.

Problem 12: Choose two new piles and make a third that balances them. Give the game to a partner, then restore balance after their move. Record your start and one pair of moves.